problem is converted to a linear one by an appropriate transform.

**Huffman encoding:** Data compression method that assigns frequently encountered characters fewer bits than seldom used characters.

**Hyperspace:** Term used in target detection and neural network analysis. One parameter can be graphically interpreted as a *line*, two parameters a *plane*, three parameters a *space*, and more than three parameters a *hyperspace*.

**Imaginary part:** The portion of a complex number that has a $j$ term, such as 2 in $3 + 2j$. In the real Fourier transform, the *imaginary part* also refers to the portion of the frequency domain that holds the amplitudes of the sine waves, even though $j$ terms are not used.

**Impulse:** A signal composed of all zeros except for a very brief pulse. For discrete signals, the pulse consists of a single nonzero sample. For continuous signals, the width of the pulse must be much shorter than the inherent response of any system the signal is used with.

**Impulse decomposition:** Breaking an $N$ point signal into $N$ signals, each containing a single sample from the original signal, with all the other samples being zero. This is the basis of convolution.

**Impulse response:** The output of a system when the input is a normalized impulse (a delta function).

**Impulse train:** A signal consisting of a series of equally spaced impulses.

**Independent variable:** In a signal, the dependent variable depends on the value of the independent variable. Example: when a voltage changes over time, time is the independent variable and voltage is the dependent variable.

**Infinite impulse response (IIR):** An impulse response that has an infinite number of nonzero values, such as a decaying exponential. Often used to indicate that a filter is carried out by using recursion, rather than convolution.

**Integers:** Whole numbers: $\cdots -2, -1, 0, 1, 2, \cdots$. Also refers to numbers stored in fixed point notation. See *floating point* for comparison.

**Interlaced decomposition:** Breaking a signal into its even numbered and odd numbered samples. Used in the FFT.

**Interlaced video:** A video signal that displays the even lines of each image followed by the odd lines. Used in television; developed to reduce flicker.

**Interpolation:** Increasing the sampling rate of a digitized signal. Generally done by placing zeros between the original samples and using a low-pass filter. See *decimation* for comparison.

**Inverse transform:** The synthesis equation of the Fourier transform, calculating the time domain from the frequency domain. See *forward transform* for comparison.

**Iterative:** Method of finding a solution by gradually adjusting the variables in the right direction until convergence is achieved. Used in CT reconstruction and neural networks.

**JPEG:** A common image file format using transform (lossy) compression. Widely used on the world wide web for graphics. See *GIF* and *TIFF* for comparison.

**Kernel:** The impulse response of a filter implemented by convolution. Also known as the *convolution kernel* and the *filter kernel*.

**Laplace transform:** Mathematical method of analyzing systems controlled by differential equations. A main tool in the design of electric circuits, such as analog filters. Changes a signal in the time domain into the s-domain

**Learning algorithm:** The procedure used to find a set of neural network weights based on examples of how the network should operate.

**Line pair:** Imaging term for *cycle*. For example, 5 cycles per mm is the same as 5 line pairs per mm.

**Line pair gauge:** A device used to measure the resolution of an imaging system. Contains a series of light and dark lines that move closer together at one end.

**Line spread function (LSF):** The response of an imaging system to a thin line in the input image.

**Linear phase:** A system with a phase that is a straight line. Usually important because it means the impulse response has left-to-right symmetry, making rising edges in the output signal look the same as falling edges. See also *zero phase*.

**Linear system:** By definition, a system that has the properties of additivity and homogeneity.